

RV Educational Institutions®

RV Institute of Technology and Management

(Affiliated to VTU, Belagavi)

JP Nagar 8th Phase, Bengaluru - 560076

Department of Mechanical Engineering

RENEWABLE ENERGY POWER PLANTS

(BME654B)



Course Name: RENEWABLE ENERGY POWER PLANTS

Course Code: BME654B

VI Semester

2025 Scheme

Prepared By :

Dr. Lokesh B. Paradeshi

Assistant Professor,

Department of Mechanical Engineering

RVITM, Bengaluru - 560076

RV Educational Institutions®
RV Institute of Technology and Management
(Affiliated to VTU, Belagavi)
Department of Mechanical Engineering

VISION

Constantly endeavour to disseminate quality education in Mechanical Engineering with cutting edge technology leading to quantum leap in innovations for societal needs.

MISSION

- To produce quality mechanical engineers by providing ICT based state-of-the-art facilities for experiential learning.
- To bridge the gap between the Industry and Academia to make the students industry-ready by close interaction with Industry.
- To infuse best mechanical engineering practices through quality education, ingenuity, innovation and entrepreneurial skills leading to incubation ecosystem.
- To provide best academic experience to peruse their goals of higher education and research in top universities.

PROGRAM SPECIFIC OBJECTIVES (PSOS)

- **PSO1:** Apply the knowledge of material science, design, manufacturing technology and Thermal engineering to solve industry related problems.
- **PSO2:** Proficient to use all appropriate software to solve complex engineering problems.
- **PSO3:** Impart quality knowledge in the field of management and interdisciplinary technologies to suit industry and society.

Contents

1	Biomass Power Plants and Hydrogen Energy:	3
1.1	Biomass:	3
1.1.1	Advantages	4
1.2	Factors affecting Biogas generation:	6
1.3	Photosynthesis Process	9
1.4	Biochemical reactions in Anaerobic Digestion:	10
1.5	Classification of the Biogas Plants	12
1.5.1	Single stage continuous plant	12
1.5.2	Two stage continuous plant	13
1.5.3	Indian Digester (Floating drum type/Khadi Village Industries Commission Plant (KVIC)):	14
1.5.4	Janata type biogas plant (Chinese):	16
1.5.5	Deenbandhu biogas plant	17
1.6	Problems Related to Bio-gas Plants	18
1.7	Advantages of Biogas:	19
1.8	Application of bio-gas in engines:	21
1.8.1	Biogas in Diesel Engine applications:	21
1.8.2	Biogas in Dual Fuel Engine applications:	22
1.8.3	Biogas As Alternate Fuel In Diesel Engines:	22
1.8.4	Biogas As Alternate Fuel In Diesel Engines:	23
1.8.5	Biogas in HCCI Engine applications:	23



1.9	Hydrogen energy:	24
1.9.1	Properties of Hydrogen	24
1.9.2	Production	25
1.10	Electrolysis of Water:	25
1.11	Thermal Decomposition of Water (thermolysis of water):	27
1.12	Bio photolysis:	28
1.13	Thermochemical Production	29
1.13.1	Steam Reforming:	29
1.13.2	Partial Oxidation or Ceramic Membrane Reactor:	30
1.13.3	Biomass Gasification and Pyrolysis:	30

Chapter 1

Biomass Power Plants and Hydrogen Energy:

Biomass Power Plants: Biomass as a renewable energy source: types and characteristics, Conversion technologies: combustion, gasification, and anaerobic digestion, biomass feedstock selection and availability, Environmental impacts and sustainability of biomass power plants, Integration of biomass power plants with other energy systems

Hydrogen Energy: Properties of Hydrogen with respected to its utilization as a renewable form of energy, sources of hydrogen, production of hydrogen, electrolysis of water, thermal decomposition of water, thermos Chemical production biochemical production.

1.1 Biomass:

It is the organic matter consisting of plant animal matter. Any matter which is biodegradable is known as biomass or organic matter. Generation of energy from biomass is referred to as Photo chemical 'harnessing of solar radiation since to generate biomass; solar radiation is a must as seen from the following equation

Solar radiation $\longrightarrow$ Photosynthesis $\longrightarrow$ Biomass $\longrightarrow$ Energy

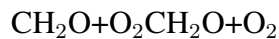
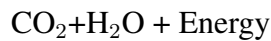
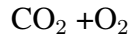
Solar radiation Photosynthesis Biomass Energy Energy from the biomass is generated in three different forms namely

i)Direct burning,



- ii) Liquefaction,
- iii) Gas generation.

i. Direct burning: When biomass is directly burnt, energy is generated as given by the following expression,



Thus when photosynthesis reaction is reversed energy is liberated.

ii. Liquefaction: Biomass is liquefied either by thermo-chemical method or biochemical method to generate alcohols like methyl and ethyl alcohol. These are mixed with petrol and used in IC Engines as fuels.

iii. Bio gas: Biomass is converted to biogas by the process of digestion or fermentation in

the presence of micro-organisms. This biogas mainly contains methane which is a good combustible gas.

Biogas consists of 50-55% of methane, 30-35% of CO₂ and remaining waste gases like H₂, N₂, H₂S etc. since it contains a hydrocarbon gas it is a very good fuel and hence can be used in IC engines. It is a slow burning gas with calorific value of 5000-5500 Kcal/kg. the raw material used to generate this are algae, crop residue, garbage, kitchen waste, paper waste, waste from sugar cane refinery, water hyacinth etc. apart from the above-mentioned raw materials excreta of cattle, piggery waste and poultry droppings are also used as raw materials.

Biogas is generated by fermentation or digestion of organic matter in the presence of aerobic and anaerobic micro-organisms. Fermentation is the process of breaking down the complex organic structure of the biomass to simple structures by the action of micro-organisms either in the presence of O₂ or in the absence of O₂. The container in which the digestion takes place is known as the digester.

1.1.1 Advantages

- The initial investment is low for the construction of biogas plant.



- The technology is very suitable for rural areas.
- Biogas is locally generated and can be easily distributed for domestic use.
- Biogas reduces the rural poor from dependence on traditional fuel sources, which lead to deforestation.
- The use of biogas in village helps in improving the sanitary condition and checks environmental pollution.
- The by-products like nitrogen rich manure can be used with advantage.
- Biogas reduces the drudgery of women and lowers incidence of eye and lung diseases.

The digestion takes place in the following steps

- i) Enzymatic hydrolysis
- ii) Acid formation
- iii) Methane formation.

i) **Enzymatic hydrolysis:** In this step the complex organic matter like starch, protein, fat, carbohydrates etc are broken down to simple structures using anaerobic micro-organisms.

ii) **Acid formation:** In this step the simple structures formed in the enzymatic

hydrolysis step are further reacted by anaerobic and facultative micro-organisms (which thrive in both the presence and absence of oxygen) to generate acids.

i) **iii. Methane formation:** In this step the organic acids formed are further converted to methane and CO₂ by anaerobic micro-organisms (anaerobes).



1.2 Factors affecting Biogas generation:

- 1) PH value
- 2) Temperature
- 3) Total solid content
- 4) Load rating
- 5) Seeding
- 6) Uniform feeding
- 7) Dia to depth ratio
- 8) Carbon to nitrogen ratio
- 9) Nutrient
- 10) Mixing
- 11) Retention time
- 12) Type of feedstock
- 13) Toxicity
- 14) Pressure

- 1) **PH value:** It is an index of hydrogen ion concentration in the mixture which also predicts acidity or alkalinity of the mixture. For effective gas generation the required PH value is 6.5 to 7.6. If this value decreases to 4-6, the mixture becomes acidic and if the value becomes 9-10 then it becomes alkaline. Both for acidic and alkaline conditions the methane forming bacteria becomes inactive and the gas generation is reduced. Thus for effective gas generation the required PH value is 6.5-7.5.
- 2) **Temperature:** The effect of temp on gas generation is as shown in graph. The two curves represent two types of bacteria which are sensitive to two different temp levels. Mesophilic type of bacteria will effectively generate gas at a temp of about 35° C. Thermophilic type of bacteria will generate gas effectively at a temp of about 55° C. As the



temperature decreases or increases from the above values the period of gas generation will be increased. Since it is easy to maintain a temp of 35° C, it is advisable to select mesophilic type of bacteria for digestion.

- 1) **Total solid content:** The raw material added to the digester contains both solid

and liquid in the ratio of 20:80 by weight. From the experience it is found that the gas generation is improved by maintaining the solid content of the mixture at about 8 to 10% by weight. This is done by adding water to the mixture.

- 1) **Loading rate:** It is the addition of the raw material to the digester/day/unit volume.

The effective load rating is found to be 0.5 to 1.6 kg of solid material/day/m³.

- 1) **Seeding:** During digestion the methane forming bacteria are consumed rapidly and their number will decrease affecting the gas generation. In order to maintain the quantity of methane forming bacteria, digested slurry from the previous batch is added to the digester. The digested slurry is rich in methane forming bacteria and the process is known as seeding.
- 2) **Uniform feeding:** this is one of the prerequisites of good digestion. The digester must be fed at the same time every day with a balanced feed of the same quality and quantity.
- 3) **Dia to depth ratio:** from the experiments it is seen that the gas generation is improved by maintaining a dia to depth ratio of 0.66 to 1. This provides uniform temp distribution throughout the digester resulting in increased gas generation.
- 4) **Carbon to nitrogen ratio:** The bacteria in the digester utilize carbon for energy

generation (as food) while nitrogen is used for cell building. Hence a carbon to nitrogen ratio of 30:1 is maintained for effective gas generation. If the ratio is not maintained the availability of carbon and nitrogen will vary resulting in reduced gas generation.

- 1) **Nutrients:** The nutrients required by the bacteria for food digestion are hydrogen, nitrogen, oxygen, carbon, phosphorous and Sulphur. Of these nutrients and



phosphorous have to be provided externally while the others are contained in the raw material itself. Nitrogen is provided by adding leguminous plants (plants with seeds enclosed in casings, eg: Maize) which are rich in nitrogen content. Phosphorous is provided by adding night soil (soil mixed with excreta of animals and humans) to the digester.

1) **Mixing:** Since bacteria in the digester have very limited reach to their food it is

necessary that the slurry is properly mixed and the bacteria get their food supply. It is found that the slight mixing improves the digestion and a violent mixing retards the digestion.

1) **Retention time:** It is the time period required for the gas generation. It completely

depends on the type of the raw materials used. Eg: Night soil requires 30 days, pig dump and poultry droppings require 20 days while cow dung and other kitchen waste requires 50 days of retention time.

1) **Type of feed stock:** The usual feed stock used are cow dung, human excreta,

poultry dropping, pig dump, kitchen waste etc. To obtain an efficient digestion these feed stocks are in some proportions, Predigested and finally chopping will be helpful for fibrous type of raw materials.

1) **Toxicity:** If the digester is left with the digested slurry it results in toxicity which

in turn reduces the gas generation. Hence the digested slurry should be removed after the gas is generated.

1) **Pressure:** It is found that the gas generation is increased with the decrease in the

pressure of the digester.

1.3 Photosynthesis Process

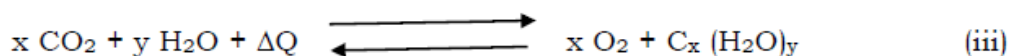
Solar radiation incident on green plants and other photosynthetic organisms performs two basic functions: (i) temperature control for chemical reactions to proceed, and (ii) Photosynthesis process. The fundamental conversion process in green plants is photosynthesis, which is the process of combining CO₂ from the atmosphere with water plus light energy to produce oxygen and carbohydrates (sugars, starches, celluloses and hemicelluloses). They are the ultimate source of most of our foods and other necessities of daily life such as clothes (in the form of cotton), furniture (in the form of wood), etc.



The generalized symbol C_x (H₂O)_y is used to indicate the carbohydrates. The products of this reaction are about 5 eV per C atom higher in energy than the initial material. Photosynthesis is a complex process. It involves several successive stages, but the overall basic Reaction is the formation of hexose (glucose, fructose, etc.) as represented by:



More complex hydrocarbons (sucrose, starch, cellulose, etc.) are formed by a chain of these simple structures. The reverse of this process is called respiration, in which CO₂, H₂O and energy are produced using carbohydrate and oxygen. In green plants, both photosynthesis and respiration occur during the day and only respiration at night. This is shown in figure below. There is a net overall gain of energy in the process, as the rate of energy loss in respiration is much less as compared to rate of energy gain during photosynthesis process. The process also results in net gain of oxygen and fixation of carbon in the form of biomass. The net energy absorbed from solar radiation during photosynthesis can be measured from its combustion.



ΔQ is enthalpy change of the combustion process, equal to the energy absorbed from photons of solar radiation, less the energy of respiration during growth. The value of ΔQ is 4.8eV per carbon atom, 470 kJ per mole of carbon or 16 MJ/kg of dry carbohydrate material. It is to be noted that the combustion requires the temperature of approximately 4000C, whereas respiration occurs at 200C through catalytic enzyme reactions.

The uptake of CO₂ by a plant leaf is a function of many factors, especially temperature, CO₂ concentration and the intensity and wavelength distribution of light. Solar radiation incident on a leaf is reflected, transmitted and absorbed. Part of the absorbed radiation (<5%) provides the energy stored in the photosynthesis and produces oxygen and carbohydrate; the remainder is absorbed in the plant as sensible heat raising its temperature, or as latent heat for water evaporation. Absorption is usually most marked in the blue and red regions.

1.4 Biochemical reactions in Anaerobic Digestion:

There are four key biological and chemical stages of anaerobic digestion ??:

- i. Hydrolysis
- ii. Acido genesis
- iii. Aceto genesis
- iv. Methano genesis.

There are four key biological and chemical stages of anaerobic digestion: Hydrolysis
Acido genesis Aceto genesis Methano genesis.

In most cases biomass is made up of large organic compounds. In order for the microorganisms in anaerobic digesters to access the chemical energy potential of the organic material, the organic matter macromolecular chains must first be broken down into their smaller constituent parts. These constituent parts or monomers such as sugars are readily available to microorganisms for further processing. The process of breaking these chains and dissolving the smaller molecules into solution is called hydrolysis. Therefore hydrolysis of high molecular weight molecules is the necessary first step in an aerobic digestion. It may be enhanced by mechanical, thermal or chemical pretreatment of the

waste. Hydrolysis step can be merely biological (using hydrolytic microorganisms) or combined: bio-chemical (using extra cellular enzymes), chemical (using catalytic reactions) as well as physical (using thermal energy and pressure) in nature.

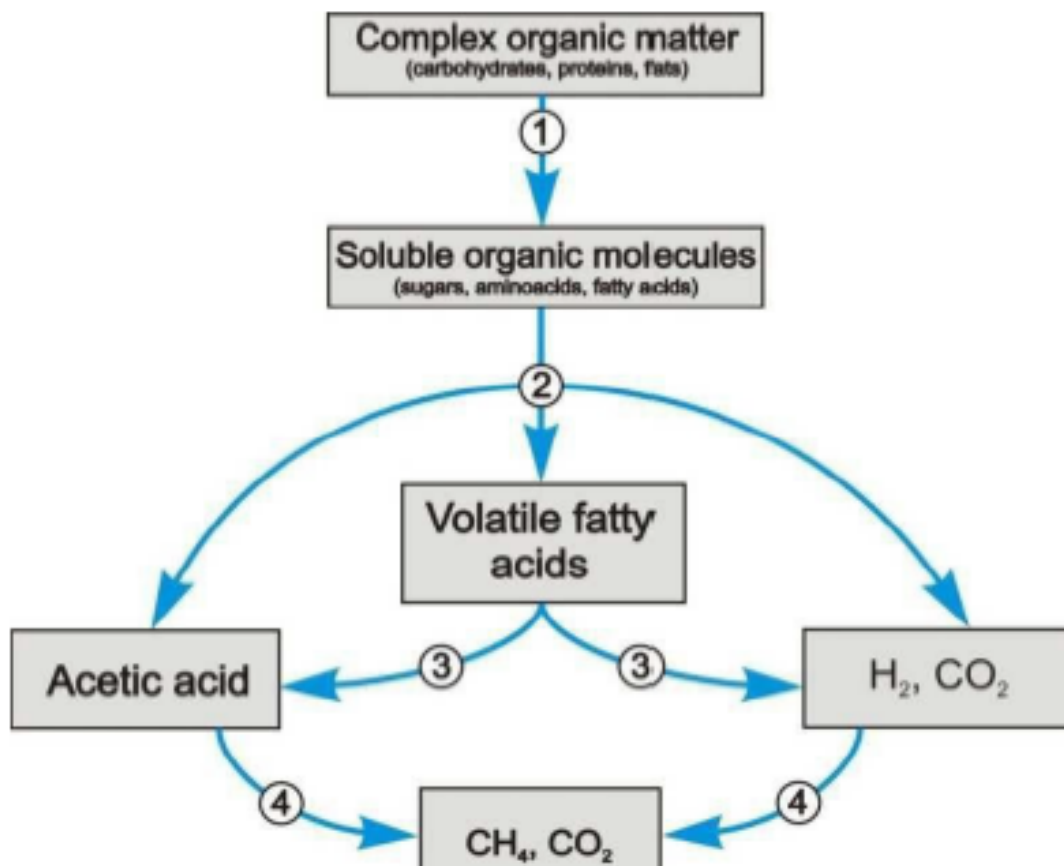


Fig. 1.1 Anaerobic pathway of complex organic matter degradation

Acetates and hydrogen produced in the first stages can be used directly by methanogens. Other molecules such as volatile fatty acids (VFA's) with a chain length that is greater than acetate must first be catabolised into compounds that can be directly utilized by methanogens. The biological process of acidogenesis is where there is further break down of the remaining components by acidogenic (fermentative) bacteria. Here VFA's are generated along with ammonia, carbondioxide and hydrogensulphide as well as other by-products.

The third stage anaerobic digestion is acetogenesis. Here simple molecules created through the acidogenesis phase are further digested by acetogens to produce largely acetic acid (or its salts) as well as carbon dioxide and hydrogen.

The final stage of anaerobic digestion is the biological process of methanogenesis. Here methanogenic archaea utilize the intermediate products of the preceding stages and convert them to methane, carbon dioxide and water. It is these components that make up the majority of the biogas released from the system. Methanogenesis is—beside other factors—sensitive to both high and low pH values and performs well between pH 6.5 and pH 8. The remaining, non-digestible organic and mineral material, which the microbes cannot feed upon, along with any dead bacterial residues constitutes the solid digestate.

1.5 Classification of the Biogas Plants

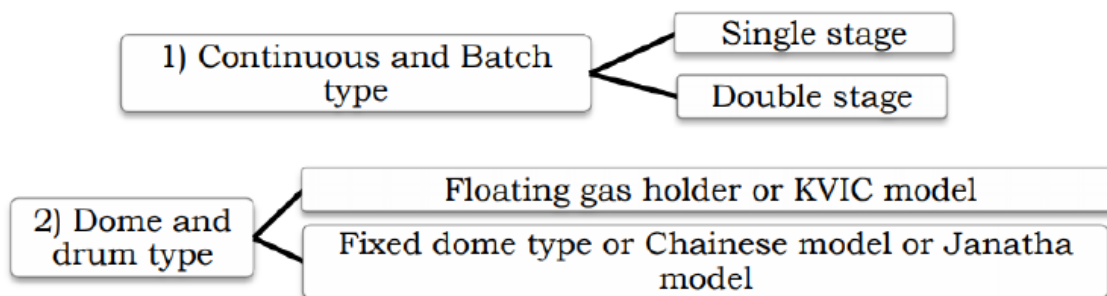


Fig. 1.2 Classification of Bio gas Plants

1.5.1 Single stage continuous plant

The entire process of conversion of complex organic compounds into biogas is completed in a single chamber. This chamber is regularly fed with raw materials while the spent residue keeps moving out. Serious problems are encountered with agricultural residues when fermented in a single stage continuous process.

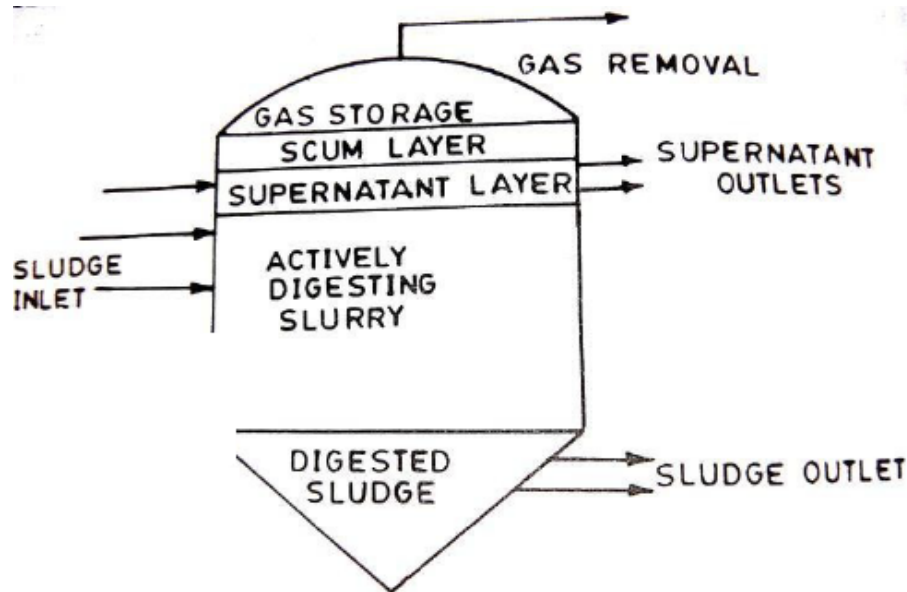


Fig. 1.3 Single stage continuous plant

1.5.2 Two stage continuous plant

The acidogenic stage and methanogenic stage are physically separated into two chambers. Thus the first stage of acid production is carried out in a separate chamber and only the diluted acids are fed into the second chamber where bio- methanation takes place and the biogas can be collected from the second chamber. Considering the problems encountered in fermenting fibrous plant

waste materials the two-stage process may offer higher potential of success. However, appropriate technology suiting to rural India is needed to be developed based on the batch stage process.

The main features of continuous plant are that:

- 1) It will produce gas continuously;
- 2) It requires small digestion chambers;
- 3) It needs lesser period for digestion;
- 4) It has fewer problems compared to batch type and it is easier in operation.

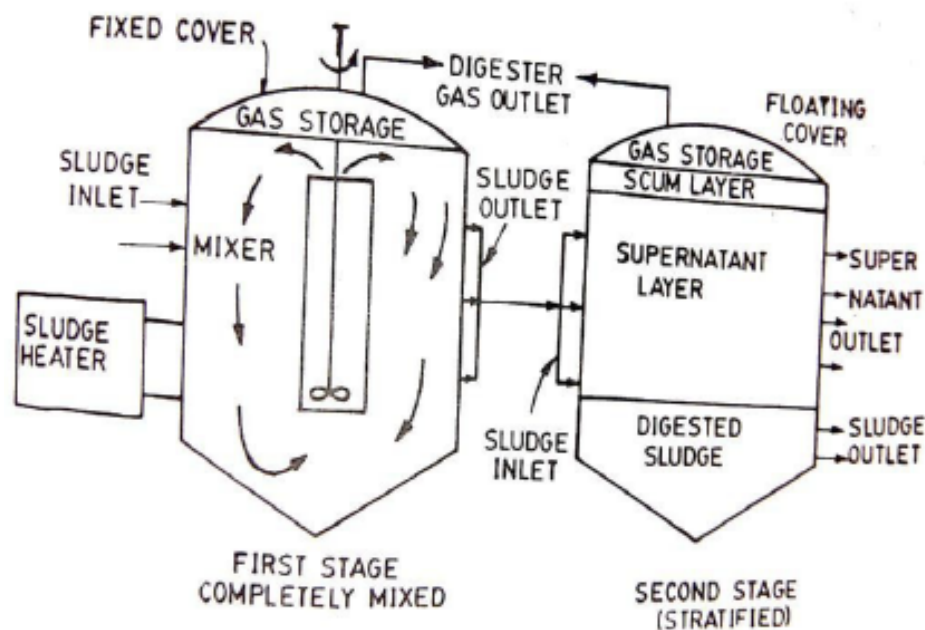


Fig. 1.4 Two stage continuous plant

1.5.3 Indian Digester (Floating drum type/Khadi Village Industries Commission Plant (KVIC)):

This mainly consists of a digester or pit for fermentation and a floating drum for the collection of gas (1.5). Digester is 3.5-6.5 m in depth and 1.2 to 1.6 m in diameter. There is a partition wall in the center, which divides the digester vertically and submerges in the slurry when it is full. The digester is connected to the inlet and outlet by two pipes. Through the inlet, the dung is mixed with water (4:5) and loaded into the digester. The fermented material will flow out through outlet pipe. The outlet is generally connected to a compost pit. The gas generation takes place slowly and in two stages. In the first stage, the complex, organic substances contained in the waste are acted upon by a certain kind of bacteria, called acid formers and broken up into small-chain simple acids. In the second stage, these acids are acted upon by another kind of bacteria, called methane formers and produce methane and carbon dioxide.

Gas holder:

The gas holder is a drum constructed of mild steel sheets. This is cylindrical in shape with concave top. The top is supported radially with angular iron stripes. The holder fits into the digester like a stopper. It sinks into the slurry due to its own weight and rests upon the ring constructed for this purpose. When gas is generated the holder rises and floats freely on the surface of slurry. A central guide pipe is provided to prevent the holder from tilting. The holder also acts as a seal for the gas. The gas pressure varies between 7 and 9 cm of water column. Under shallow water table conditions, the adopted diameter of digester is more and depth is reduced. The cost of drum is about 40% of total cost of plant. It requires periodical maintenance. The unit cost of KVIC model with a capacity of 2 m³/day costs approximately Rs.14, 000/.

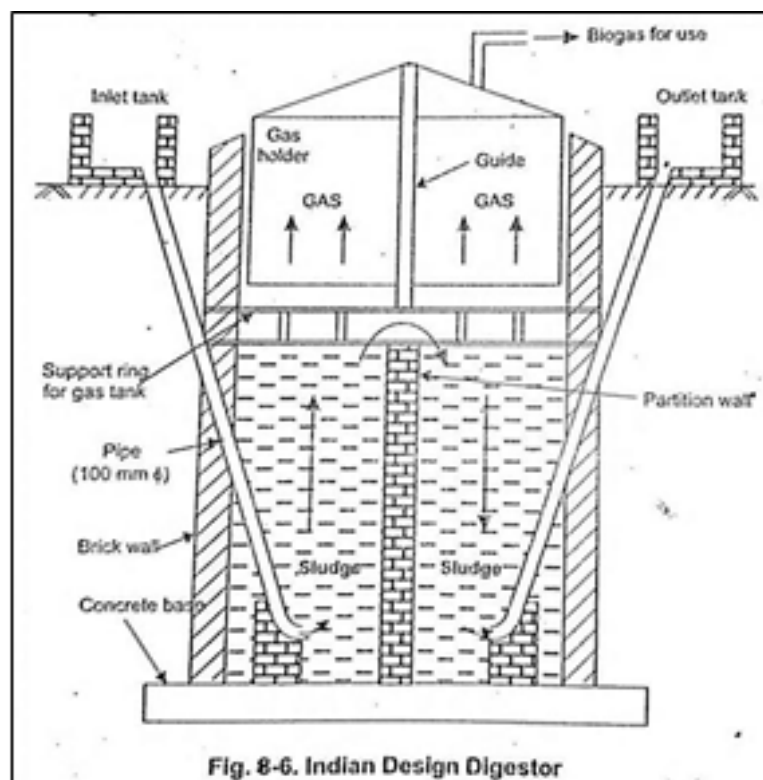


Fig. 1.5 Indian Digester (Floating drum type/Khadi Village Industries Commission Plant (KVIC))

1.5.4 Janata type biogas plant (Chinese):

The design of this plant is of Chinese origin but it has been introduced under the name —Janata biogas plant (1.6) by Gobar Gas Research Station, Ajitmal in view of its reduced cost. This is a plant where no steel is used, there is no moving part in it and maintenance cost is low. The plant can be constructed by village mason taking some pre-explained precautions and using all the indigenously available building materials. Good quality of bricks and cement should be used to avoid the afterward structural problems like cracking of the dome and leakage of gas. This model has a higher capacity when compared with KVIC model, hence it can be used as a community

biogas plant. This design has longer life than KVIC models. Substrates other than cattle dung such as municipal waste and plant residues can also be used in Janata type plants. The plant consists of an underground well sort of digester made of bricks and cement having a dome-shaped roof which remains below the ground level as shown in figure. At almost middle of the digester, there are two rectangular openings facing each other and coming up to a little above the ground level, act as an inlet and outlet of the plant. Dome shaped roof is fitted with a pipe at its top which is the gas outlet of the plant.

The principle of gas production is same as that of KVIC model. The biogas is collected in the restricted space of the fixed dome; hence the pressure of gas is much higher, which is around 90 cm of water column.

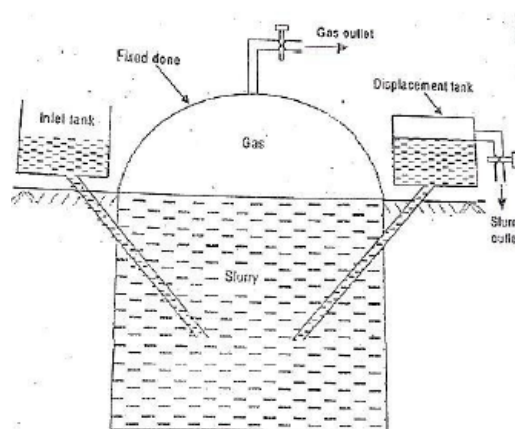


Fig. 1.6 Janata type biogas plant (Chinese):

1.5.5 Deenbandhu biogas plant

Deenbandhu model (1.7) was developed in 1984, by Action for Food Production (AF-PRO), a voluntary organization based in New Delhi. Schematic diagram of a Deenbandhu biogas plant entire biogas programme of India as it reduced the cost of the plant half of that of KVIC model and brought biogas technology within the reach of even the poorer sections of the population. The cost reduction has been achieved by minimizing the surface area through joining the segments of two spheres of different diameters at their bases. The cost of a Deenbandhu plant having a capacity of 2 m³/day is about Rs.8000/-. The Deenbandhu biogas plant has a hemispherical fixed- dome type of gas holder, unlike the floating dome of the KVIC-design is shown. The dome is made from pre-fabricated Ferro cement or reinforced concrete and attached to the digester, which has a curved bottom. The slurry is fed from a mixing tank through an inlet pipe connected to the digester. After fermentation, the biogas collects in the space under the dome. It is taken out for use through a pipe connected to the top of the dome, while the sludge, which is a by-product, comes out through an opening in the side of the digester. About 90% of the biogas plants in India are of the Deenbandhu type.

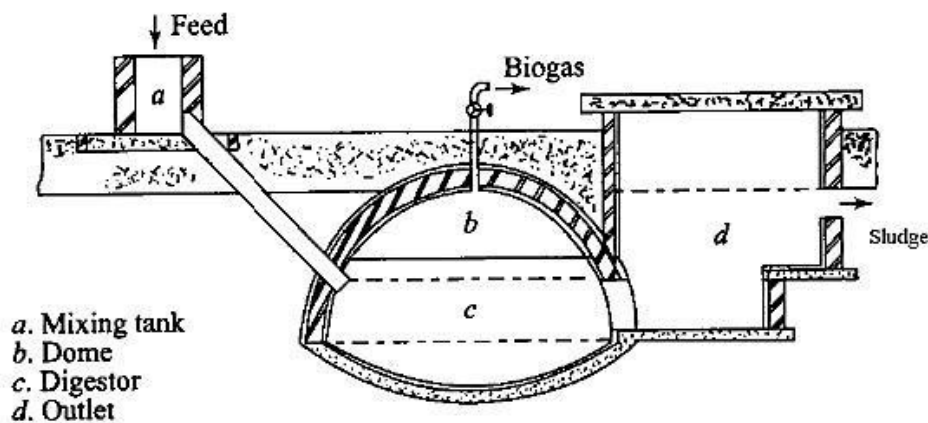


Fig. 1.7 Janata type biogas plant (Chinese):

1.6 Problems Related to Bio-gas Plants

Some problems are natural and some are created by the persons biogas plants owners but all are controllable.

1. Handling of effluent slurry is major problem if the person is not having sufficient open space or compost pits to get the slurry dry. Use of press filters and transportation is expensive and out of reach of poor farmers. For a domestic plant, 200 litres capacity oil drums can be used to carry this effluent to the fields but this will require some human-animal labour or consumption of diesel if a auto vehicle is used.
2. The gas forming-methanogenic bacteria are very sensitive towards the temperature compared to those of non-methanogenic. During winter as the temperature falls, there is decrease in the activity of methanogenic bacteria and subsequently fall in gas production rate. Many methods have been suggested to overcome this temperature problem as described earlier, e.g.,
 - a. Use of solar heated hot water to make a slurry of influent but the temperature of water should not exceed 60oC otherwise the mesophilicbacteria will die.
 - b. Circulation of hot water obtained either from solar heater or I.C. engine heat exchanger, through pipes inside the digester.
 - c. Green house effect also give good results but it is costlier and after few years the polythene sheet used in it becomes opaque.
 - d. Addition of various nutrients for bacteria.
 - e. Converting the biogas plant by straw bags during night hours.
3. Due to lack of proper training to the bio-gas plant owners for the operation of plant, a lot of problems arises. It has been noticed that many persons increase the loading rate and some also do not try to mix the cattle dung with water, keeping in mind more gas production. Due to this, the flowof slurry from inlet towards outlet is very slow or even stops. This maycause accumulation of volatile fatty acids and drop in Ph and then failure of digester. Also it is not possible to stir the digester content of high solid concentration.
4. Some persons add urea-fertilizer in large quantities due to which toxicity of ammonia nitrogen may cause a decrease in gas production.



5. pH and volatile fatty acids play an important role in anaerobic digestion and should remain under optimum range otherwise this may cause upsetting of digester and even its failure. pH can be checked from time to time by the use of cheap and easily available pH paper but volatile fatty acids can only be determined in a laboratory having its testing facilities. For controlling pH in optimum range, it tends to fall below 7.0, lime has been suggested, as it is easily available cheap material and does not harm the activity of bacteria.
6. Leakage of gas from gas holder especially in case of Janta type biogas plants is a major and very common problem. When there is quite enough gas in a gas holder, the leakage should be checked by using water and the points marked and then get repaired. During repairing there should be no gas inside the gas holder and the stop cock remains open till repaired points get dry. Quality of constructing material such as cement is important.

1.7 Advantages of Biogas:

1. Biogas is an energy carrier which can be used for several energy applications (eg. electricity generation, heat production, combined heat and power production, transport fuel, injection to the natural gas grid).
2. Biogas can contribute to several sectors:
 - i) Environment
(eg. Fight against Climate change)
 - i) Energy
(eg. Energy security, local source)
 - i) Agriculture
(eg. Sustainable cultivation and animal breeding)
 - i) Society



(eg. Employment enhancement, rural development)

1. Some Environmental benefits of biogas:

- i) Reduced emissions of greenhouse gases, direct and indirect (eg. CO₂, CH₄ and nitrous oxide –N₂O).
- ii) Water and Waste management (Reduced consumption of resources and increased recycling, reduced water environment pollution from leaching of nutrients, environmental friendly solution to the

waste disposal problem).

- i) Reduced odour and flies nuisances.
- ii) Soil and landscape

1. Emissions reduction of greenhouse gases (eg. CO₂, CH₄ and nitrous oxide –N₂O).

Direct: The combustion of biogas also releases CO₂. Compared to fossil fuels, the carbon in biogas was recently up taken from the atmosphere, by photosynthetic activity of the plants (closed carbon cycle) .

- Biogas production by AD reduces also emissions of methane (CH₄) and nitrous oxide (N₂O) from storage and utilization of animal manure as fertilizer. It is worth mentioning that although biogas is a potential low
- carbon energy source, this depends on the way biogas is produced. In the case that biogas comes from residues, waste or from energy crops grown on abandoned agricultural and this offers sustained GHG advantages.
- Emissions reduction of greenhouse gases (eg. CO₂, CH₄ and nitrous oxide –N₂O). Indirect:
- Utilization of biogas substitute fossil fuel (such as lignite, hard coal, crude oil and natural gas) and thus reduces emissions (externalities).

1. Water and Waste management



- Compared to other biofuels, biogas needs the lowest amount of process water. This aspect is very important since many regions of the world face huge water problems
 - One of the main advantages of biogas production is the ability to transform waste material into a valuable resource, by using it as feedstock for AD.
 - Biogas technologies contribute to reduce the volume of wastes and the costs for waste disposal (transportation, disposal).
1. Health issues, odor and flies utilization of digestate as fertilizer improves veterinary safety, when compared to untreated manure and slurries.

AD reduces odours (positive change in the composition of odours as well). Digestate is almost odourless and the remaining disappear shortly after application as fertilizer on the fields (ammonia odours).

1. Employment
2. Rural development
3. Local economy and Energy Market development

1.8 Application of bio-gas in engines:

1.8.1 Biogas in Diesel Engine applications:

Biogas generally has a high self-ignition temperature hence; it cannot be directly used in a CI engine. So it is useful in dual fuel engines. The dual fuel engine is a modified diesel engine in which usually a gaseous fuel called the primary fuel is inducted with air into the engine cylinder. This fuel and air mixture does not auto ignite due to high octane number. A small amount of diesel, usually called pilot fuel is injected for promoting combustion. The primary fuel in dual fuelling system is homogeneously mixed with air that leads to very low level of smoke. Dual fuel engine can use a wide variety of primary and pilot fuels. The pilot fuels are generally of high cetane fuel. Biogas can also be used in dual fuel mode with vegetable oils as pilot fuels in diesel engines. Introduction

of biogas normally leads to deterioration in performance and emission characteristics. The performance of engine depends on the amount of biogas and the pilot fuel used. Measures like addition of hydrogen, LPG, removal of CO₂ etc. have shown significant improvements in the performance of biogas dual fuel engines. The ignition delay of the pilot fuel generally increases with the introduction of biogas and this will lead to advance the injection timing. Injectors opening pressure and rate of injection also are found to play important role in the case of biogas fuelled engine, where vegetable oil is used as a pilot fuel. The percentage in biogas acts as diluents to slow down the combustion process in Homogenous charged compression ignition (HCCI) engines. However, it also affects ignition. Thus a fuel with low self-ignition temperature could be used along with biogas to help its ignition. This kind of engine has shown a superior performance as compared to a dual fuel mode of operation.

1.8.2 Biogas in Dual Fuel Engine applications:

In this case, the normal diesel fuel injection system still supplies a certain amount of diesel fuel. The engine however sucks and compresses a mixture of air and biogas fuel which has been prepared in external mixing device. The mixture is then ignited by and together with the diesel fuel sprayed in. The amount of diesel fuel needed for sufficient ignition is between 10% and 20% of the amount needed for operation on diesel fuel alone. Operation of the engine at partial load requires reduction of the biogas supply by means of a gas *control* valve. A simultaneous reduction of airflow would reduce power and efficiency because of reduction of compression pressure and main effective pressure. So, the air/fuel ratio is changed by different amounts of injected biogas. All other parameters and elements of diesel engine remain unchanged.

1.8.3 Biogas As Alternate Fuel In Diesel Engines:

A Literature Review Modification of diesel engine into dual fuel engine Advantages- Operation on diesel fuel alone is possible when biogas is not available.-Any contribution of biogas from 0% to 85% can substitute a corresponding part of diesel fuel while performance remains as in 100% diesel fuel operation.-Because of existence of a governor at most diesel engines automatic control of speed/power can be done by changing the

amount of diesel fuel injection while the biogas flow remains uncontrolled. Diesel fuel substitutions by biogas are less substantial in this case.

1.8.4 Biogas As Alternate Fuel In Diesel Engines:

A Literature Review Modification of diesel engine into dual fuel engine Advantages- Operation on diesel fuel alone is possible when biogas is not available.-Any contribution of biogas from 0% to 85% can substitute a corresponding part of diesel fuel while performance remains as in 100% diesel fuel operation.-Because of existence of a governor at most diesel engines automatic control of speed/power can be done by changing the amount of diesel fuel injection while the biogas flow remains uncontrolled. Diesel fuel substitutions by biogas are less substantial in this case.

Limitations:

- The dual fuel engine cannot operate without the supply of diesel fuel for ignition.
- The fuel injection jets may overheat when the diesel fuel flow is reduced to 10% or 15% of its normal flow. Larger dual fuel engines circulate extradiesel fuel through the injector for cooling.
- To what extent the fuel injection nozzle can be affected is however a question of its specific design, material and the thermal load of the engine, and hence differs from case to case.
- A check of the injector nozzle after 500 hours of operation in dual fuel is recommended.

1.8.5 Biogas in HCCI Engine applications:

The Homogeneous Charge Compression Ignition (HCCI) concept is a potential for achieving a high thermal efficiency and low Nitrogen Oxide (NO) emission. The HCCI engine with 50 % biogas as a primary fuel and 50% diesel as pilot fuel gives a maximum NO of 20 ppm is a major advantage over biogas diesel dual fuel mode. In biogas diesel dual fuel mode, the presence of CO₂ in biogas lowers the thermal efficiency however, in biogas diesel HCCI (BDHCCI) mode CO₂ reduces high heat release rate. The break mean effective pressure (BMEP) in BDHCCI mode is in the range of 2.5 bar to 4 bar. The smoke and Hydro Carbon (HC) level were also low when the biogas is used as a primary fuel



for BDHCCI mode. For HCCI operation the inducted charge temperature is required to be maintained at 80-135°C, which can be obtained from the exhaust heat. Thus, biogas with HCCI engine gives high efficiency and low emission.

1.9 Hydrogen energy:

Hydrogen is the simplest element. An atom of hydrogen consists of only one proton and one electron. It is also the most plentiful element in the universe. Despite its simplicity and abundance, hydrogen doesn't occur naturally as a gas on the Earth – it is always combined with other elements. Water, for example, is a combination of hydrogen and oxygen (H₂O).

Hydrogen holds the potential to provide clean, reliable and affordable energy supply that can enhance economy, environment and security. It is flexible and can be used by all sectors of economy.

Hydrogen can be produced by using a variety of energy sources, such as solar, nuclear and fossil fuels and can be converted to useful energy forms efficiently and without detrimental environmental effects. When burned as fuel or converted to electricity it joins with oxygen to produce energy with water as the only emission. When air is used for combustion instead of oxygen, some NO_x is also produced, which can be reduced by lowering the combustion temperature.

1.9.1 Properties of Hydrogen

Hydrogen is an odorless and colorless gas. It has the simplest and lightest atom with one proton and one electron and molecular weight of 2.016. Important properties are listed and compared with natural gas and gasoline in Table below.

SN	Properties	Gasoline	Natural gas	Hydrogen
1	Density (kg/m^3)	730	0.78	0.0837
2	Boiling point, ($^{\circ}\text{C}$)	38 to 204	-156	-253 (20.3 K)
3	Lower heating value, (MJ/kg)	44.5	48	125
	(MJ/ m^3)	32	37.3	10.4 (gas), 8520 (liquid)
4	Higher heating value, (MJ/kg)	50.8	55	141.90
	(MJ/ m^3)	36.6	42.6	11.89 (gas), 10046 (liquid)
5	Flammable limit, % in air	1.4–7.6	5–16	4–75
6	Flame speed, (m/s)	0.4	0.41	3.45
7	Flame temperature, ($^{\circ}\text{C}$)	2197	1875	2045
8	Flame luminosity	High	Medium	Low

1.9.2 Production

Although hydrogen is the third most abundant element on the earth, it does not exist in Free State, except for small quantities in the upper atmosphere. It is, therefore, not a primary energy source. It can therefore, be produced through two routes:

- Fossil fuels, such as natural gas, coal, methanol, gasoline etc., and biomass are decomposed by thermo-chemical (steam reforming or partial oxidation) methods to obtain hydrogen. The CO produced in the process is eliminated by water gas shift reaction. This route of hydrogen production causes CO₂ emission. The energy content of the produced hydrogen is less than the energy content of the original fuel, some of it being lost as excessive heat during production.
- Hydrogen can also be produced by splitting water into hydrogen and oxygen by using energy from nuclear or renewable sources such as solar, wind, geothermal, etc., through electrical or thermal means (i.e. electrolysis and thermolysis respectively). Water splitting is also possible through biophotolysis process using solar radiation.

1.10 Electrolysis of Water:

Electrolysis (1.8) is the simplest method of hydrogen production. Currently, this method is not as efficient or cost effective as thermo-chemical method using fossil fuels or biomass.

But it would allow for more distributed hydrogen generation and open the possibilities for use of electricity generated from renewable and nuclear resources for hydrogen production.

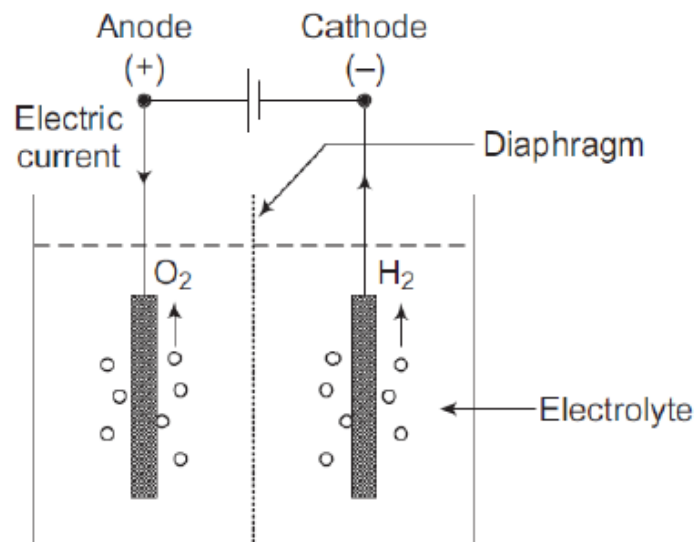


Fig. 1.8 Electrolytic cell

An electrolysis cell essentially consists of two electrodes, commonly flat metal or carbon plates, immersed in an aqueous conducting solution called electrolyte, as shown in Fig. A direct current decomposes water into H_2 and O_2 , which are released at cathode (–ve electrode) and anode (+ve electrode) respectively. As water itself is poor conductor of electricity, an electrolyte, commonly aqueous KOH is used.

Ideally, a decomposition voltage of 1.23 V per cell should be sufficient at normal temperature and pressure; however, due to various reasons a voltage of about 2 V per cell is applied in practice. The energy required is 3.9–4.6 kWh per m^3 of hydrogen produced. About 60–70 per cent of this energy is actually utilized in electrolysis.

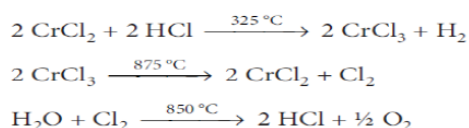
Therefore, the efficiency of electrolysis process is about 60–70 per cent, which can be improved up to 80 per cent by using catalyst such as porous platinum or nickel. A diaphragm (usually woven asbestos) prevents electronic contact between the electrodes and passage of gas or gas bubbles. Electrolysis method is most suitable when primary energy is available as electrical energy, e.g. solar photovoltaic energy. It is also suitable where cheap electricity is available from other sources such as wind, geothermal, etc.

1.11 Thermal Decomposition of Water (thermolysis of water):

When primary energy is available in the form of heat (e.g. solar thermal), it is more logical to produce hydrogen by splitting water directly from heat energy using thermolysis. This would be more efficient than conversion of heat, first to electricity (using heat engine generator) and then producing hydrogen through electrolysis. The efficiency of thermal plant is usually in range 32–38 per cent and that of electrolysis is 80 per cent. The overall efficiency through thermal-electrical-hydrogen route would thus be only 25–30 per cent.

Direct thermal decomposition of water is possible but it requires a temperature of at least 2500 °C; because of temperature limitations of conversion process equipment, direct single-step water decomposition cannot be achieved. However, sequential chemical reactions at substantially lower temperature can be devised to split water into H₂ and O₂. In the reaction series, water is taken up at one stage and H₂ and O₂ are produced in different stages. The energy is supplied as heat at one or more stages and partly released at some stage in the cycle.

Apart from decomposition of water, all other materials are recovered when the cycle is completed. Therefore, the method is known as thermo-chemical cycle. The efficiency of conversion from heat energy to hydrogen is better than its conversion through electrolysis route only when the upper temperature of thermo-chemical cycle is above 700 °C. For the upper temperature of 950 °C the efficiency of conversion is about 50 per cent. This is a marked improvement over what is possible through electrolysis route. Several thermo-chemical cycles have been proposed and are under investigation. One such cycle is given below: At present, no commercial process for thermal splitting of water using thermo chemical cycle is in operation.



At present, no commercial process for thermal splitting of water using thermo chemical cycle is in operation.

1.12 Bio photolysis:

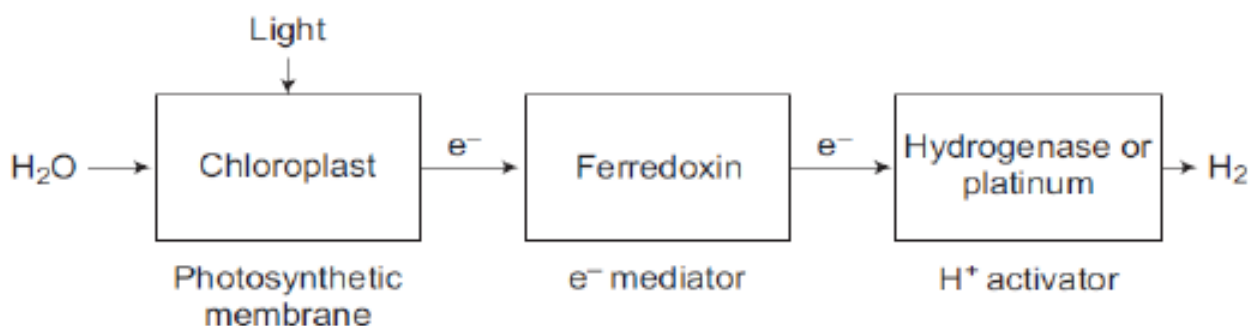


Fig. 1.9 Bio photolysis:

In this method the ability of the plants (especially algae) to split water during photosynthesis process is utilized. An artificial system is devised, which could produce hydrogen and oxygen from water in sunlight using isolated photosynthetic membrane and other catalysts. Since this process is essentially a decomposition of water using photons in the presence of biological catalysts, the reaction is called photolysis of water. There are three distinct functional components coupled together in the system as shown in Fig. 1.9

- (i) Photosynthetic membrane, which absorbs light, split water to generate oxygen, electrons and protons,
- (ii) An electron mediator, which is reducible by photo-synthetically generated electrons and
- (iii) a proton activator that will accept electrons from the reduced mediator and catalyze the reaction:



A system with chloroplast (small bodies containing the chlorophyll in green plants) as a photosynthetic membrane to split hydrogen and oxygen, ferredoxin as e⁻ mediator and hydrogen as (an enzyme) or finely dispersed platinum as proton activator, has been successfully tested. The method is being extensively studied further.

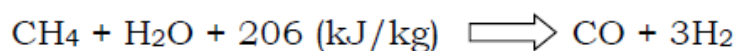
1.13 Thermochemical Production

Hydrogen bound in organic matter and in water makes up 70% of the earth's surface. Breaking up these bonds in water allows us produce hydrogen, and then, to use it as a fuel. There are numerous processes that can be used to break these bonds.

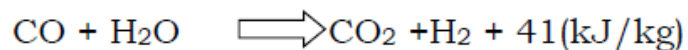
1.13.1 Steam Reforming:

Steam reforming uses thermal energy to separate hydrogen from the carbon components in methane and methanol and involves the reaction of these fuels with steam on catalytic surfaces. The first step of the reaction decomposes the fuel into hydrogen and carbon monoxide. Then, a 'shift reaction' changes the carbon monoxide and water to carbon dioxide and hydrogen. These reactions occur at temperatures of 200⁰C or greater. Steam reforming of natural gas is currently the least expensive method and is responsible for more than 90% of hydrogen production worldwide. Natural gas is first cleared from sulphur compounds. It is then mixed with steam and send over a nickel– alumina catalyst inside a tubular reactor heated externally, where carbon monoxide (CO) and hydrogen (H₂) are generated. This step is followed by a catalytic water-gas shift reaction that converts the CO and water to hydrogen and carbon dioxide (CO₂). The hydrogen gas is then purified.

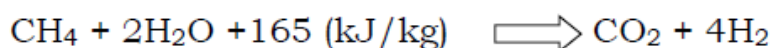
The endothermic reforming reaction is,



It is usually followed by the exothermic shift reaction:



The overall reaction is:



1.13.2 Partial Oxidation or Ceramic Membrane Reactor:

In the partial oxidation process, natural gas (or other liquid or gaseous hydrocarbons) and oxygen are injected into a high-pressure reactor. The oxygen to carbon ratio is optimally set for maximizing the yield of CO and H₂ and avoiding the formation of soot. Further steps and equipment remove the large amount of heat generated by the oxidation reaction, shift the CO with water to CO₂ and H₂, and remove the CO₂, which can then be captured, and purify the hydrogen produced. This process needs oxygen, which is usually provided by an air distillation plant. Partial oxidation can also be helped by an oxidation catalyst. It is then called catalytic partial oxidation.

The partial oxidation reaction for natural gas is:



1.13.3 Biomass Gasification and Pyrolysis:

The thermal processing techniques for plant material (biomass) and fossil fuels are similar, with a number of the downstream unit operations being essentially the same for both feedstocks. Using agricultural residues and wastes, or biomass specifically grown for energy uses, hydrogen can be produced via pyrolysis or gasification. Biomass pyrolysis produces a liquid product (bio-oil) that, like petroleum, contains a wide spectrum of components that can be separated into valuable chemicals and fuels.

Unlike petroleum, bio-oil contains a significant number of highly reactive oxygenated components derived mainly from constitutive carbohydrates and lignin. These components can be transformed into products, including hydrogen. Co-product strategies are designed to produce high value chemicals, like phenolic resins, in conjunction with hydrogen.